



BIOLOGY SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

Time allowed: 20 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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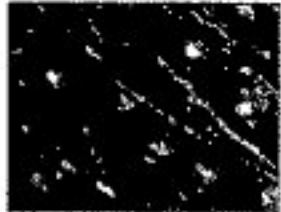
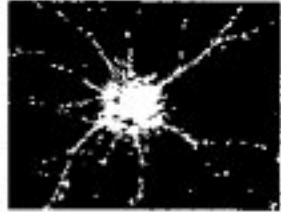
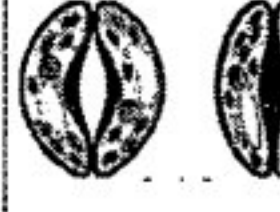
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Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Climate change research involves biology, meteorology and geology mainly to:	Reduce efficiency	Avoid experiments	Integrate different expertise	Create conflict	☐	☐	☐	☐
2. The forelimbs of human, whale, and bat provide evidence of evolution because they are:	Analogous organs	Vestigial organs	Modified organs	Homologous organs	☐	☐	☐	☐
3. If guanine is present on one strand of DNA, the base present on the complementary strand:	Adenine	Thymine	Uracil	Cytosine	☐	☐	☐	☐
4. Which condition is essential for organisms to be placed in the same species?	Can interbreed artificially	Have similar habitat	Produce fertile offspring	Produce identical offspring	☐	☐	☐	☐
5. During which phase of the cell cycle does DNA replication occur?	G ₁	G ₂	M	S	☐	☐	☐	☐
6. The smooth movement and mixing of food in the stomach is due to:	Muscular tissue	Connective tissue	Nervous tissue	Epithelial tissue	☐	☐	☐	☐
7. Mainly enzymes are made up of:	Lipids	Proteins	Nucleic acids	Carbohydrates	☐	☐	☐	☐
8. Plants obtain mineral nutrients from soil mainly through:	Root hair	Xylem cells	Parenchyma cells	Phloem cells	☐	☐	☐	☐
9. Transpiration rate decreases in high humidity because:	Air pressure rises	Stomata close	Air is saturated	Leaf area decreases	☐	☐	☐	☐
10. Which of the following processes involves cloning?	Grafting	Tissue culture	Parthenogenesis	Budding	☐	☐	☐	☐
11. After double fertilization the triploid nucleus develops into:	Endosperm	Ovule	Embryo	Seed coat	☐	☐	☐	☐
12. Which cell is responsible for transport of oxygen in animals?	 Muscle cell	 Red Blood cell	 Nerve cell	 Guard cell	☐	☐	☐	☐



BIOLOGY SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

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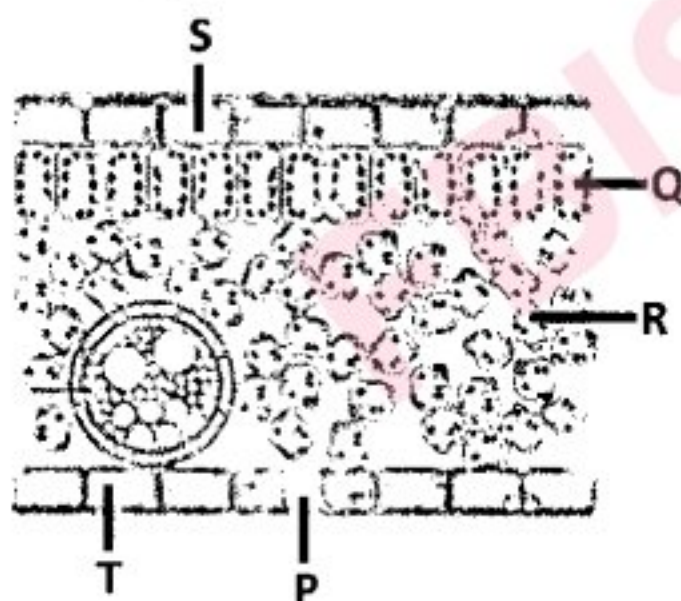
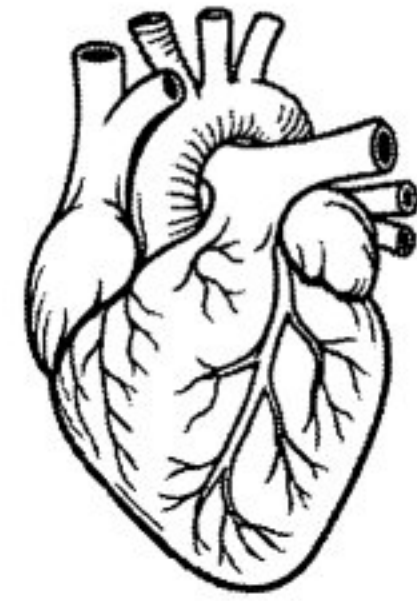
Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

Q.2 Attempt the following parts.

(10 x 3 =30)

(i)	How are different types of RNA involved in converting Hereditary information into functional proteins?	1x3	OR	Write two differences between saturated and unsaturated fatty acids. Also describe the basic structure of a lipid molecule briefly.	2+1
(ii)	Briefly explain speciation.	03	OR	Briefly explain Biotechnology and Biophysics. Give one example of each.	1.5 +1.5
(iii)	Why are starch and cellulose classified as polysaccharides? Write any two characteristics of polysaccharides.	1+2	OR	How do runners in plants help in vegetative propagation?	03
(iv)	How does temperature affect rate of transpiration?	03	OR	Name any three macronutrients required by plants along with their use in plants.	1x3
(v)	Name the domain which includes organisms without a true nucleus. Also write two characteristics of this domain.	1+2	OR	Briefly explain the role of epithelial and connective tissues in stomach.	03
(vi)	How does comparative anatomy provide support as an evidence of evolution?	03	OR	What is meant by family, genus and class. According to taxonomic ranking	1x3
(vii)	Muscle cells contain many mitochondria. How this structural feature is advantageous for their function? Justify.	03	OR	Gardeners prefer grafting to rapidly improve fruit quality. Why is this method considered as artificial propagation? Explain briefly.	03
(viii)	How does a non-competitive inhibitor affect enzyme activity?	03	OR	How does mitosis play a role in asexual reproduction and cell replacement?	1.5 +1.5
(ix)	Briefly describe Histology, Genetics and Pharmacology.	1x3	OR	Write any three differences between meiosis and mitosis.	1x3
(x)	Cross-section of a leaf is given. Identify: a. What is the role of part labelled as P? b. Differentiate between the function of Q and R. c. Label the parts S and T. 	1x3	OR	Identify the level of organization in given diagram and enlist any four examples of that level of organization. 	1+2

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Sketch and label Lysosome, Vacuole and Centriole. Also write one role of each in the cell.	2+2 +2	OR	How does Meiosis I reduce the number of chromosomes to half? (Explain the stages of Meiosis I) Also draw the diagrams of Meiosis I.	4+2
Q.4	Differentiate between types of anaerobic respiration. Highlight the significance of anaerobic respiration as well.	4+2	OR	Explain development of male gametophyte in plants. Briefly explain the parts of a seed. Also draw the lifecycle of a plant.	2+2 +2
Q.5	How is the scientific method used to solve a biological problem? Explain all steps in correct sequence with brief explanation of each involved in scientific method.	06	OR	Describe the types of cofactors of enzymes. How enzymes affect energy of activation? Explain. Draw graph as well.	3+3



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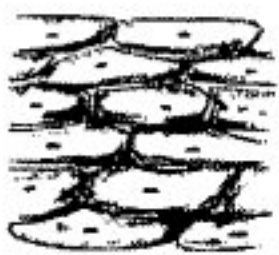
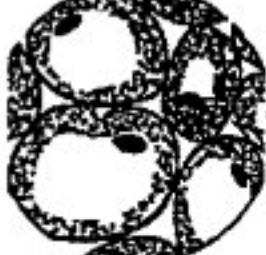
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Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. A team studying a new disease includes a biologist, computer scientist, and statistician. Which outcome best shows scientific collaboration?	Biologist works alone	Data kept secret	Shared analysis and feedback	Single opinion accepted	☐	☐	☐	☐
2. Fossils of the same organism found in different rock layers mainly show:	Sudden creation	No variation	Gradual change over time	Random destruction and blockage	☐	☐	☐	☐
3. A DNA strand having nucleotide sequence A-T-G-C-T will have complementary strand:	A-T-G-C-A	T-A-C-G-A	A-U-G-C-T	G-C-A-T-A	☐	☐	☐	☐
4. A species is best defined as a group of organisms that:	Look similar	Interbreed naturally	Live together	Share habitat	☐	☐	☐	☐
5. In G ₁ phase of cell cycle:	Nucleotides are formed	DNA replicates	Cell exits cell cycle	Cell divides	☐	☐	☐	☐
6. The main tissue in the stomach responsible for secretion of gastric juice is:	Epithelial	Nervous	Muscular	Connective	☐	☐	☐	☐
7. An enzyme is best described as a/an:	Lipid molecule	Hormone	Energy source	Biological catalyst	☐	☐	☐	☐
8. Magnesium is required for the synthesis of:	Chlorophyll	Cell wall	Guard cells	Proteins	☐	☐	☐	☐
9. Which structure is NOT related to xylem, tissue in plants?	Vessels	Sieve tubes	Tracheid	Xylem fibers	☐	☐	☐	☐
10. Cloning is the process of producing:	Different cells	Mutated cells	Identical cells	Hybrid species	☐	☐	☐	☐
11. After pollination, which process must occur for seed formation?	Germination	Dispersal	Meiosis	Fertilization	☐	☐	☐	☐
12. Which cells contain more chloroplasts for photosynthesis?	 Epidermal cells	 Sclerenchyma cells	 Parenchyma cells	 Mesophyll cells	☐	☐	☐	☐

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BIOLOGY SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

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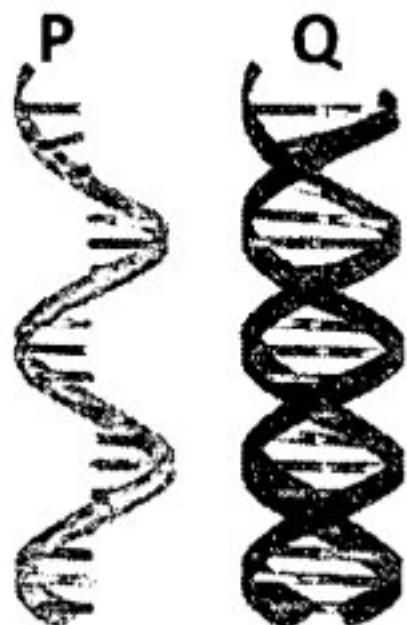
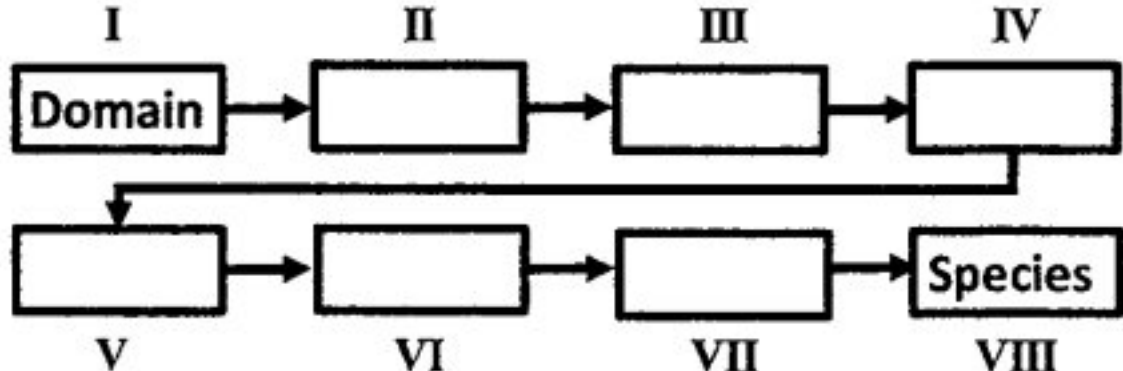
Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

Q. 2 Attempt the following parts.

(10 x 3 =30)

i)	Identify the Biomolecules P and Q. Write one function of each.		1.5 + 1.5	OR	Complete the given flowchart of Taxonomic ranks of classification in correct order. 	03
i)	Enlist main points of Darwin's theory of evolution. Briefly explain the key principle on which theory is based.	2+1	OR	Enlist any three careers related to the field of Biology and briefly state what each deals with.	1x3	
ii)	Differentiate between monosaccharides and disaccharides (any three difference).	1x3	OR	What is meant by parthenogenesis? Why is it considered a form of asexual reproduction?	1+2	
v)	How wind and humidity affect rate of transpiration?	1.5 +1.5	OR	Briefly describe macronutrients and micronutrients in plants with one example of each.	1.5 +1.5	
v)	Name and write one characteristic of three domains into which living organisms are broadly classified.	1x3	OR	Name any three basic types of tissues found in the human stomach with their functions.	1x3	
vi)	How does Paleontology support evolution?	03	OR	How is the concept of emergent properties as gain in functionality applied to going from subcellular organelles to tissues level?	03	
vii)	A plant cell placed in a hypotonic solution does not burst. How its structure provides an advantage? Give reason.	03	OR	A farmer uses stem cuttings to grow new plants instead of seeds. Identify the type of propagation and give reasons to use cuttings.	1+2	
viii)	What is competitive inhibition in enzyme action? Explain briefly.	1+2	OR	Briefly explain the role of meiosis in maintaining chromosome number in animals and plants.	1.5 +1.5	
ix)	Briefly describe Biometry and Biogeography with one example of each.	1.5 +1.5	OR	What are stem cells? Write about its types.	1+2	
x)	Name any three main tissues found in a typical leaf and state one function of each.	1x3	OR	What are lipids? Write about the basic structure of a lipid molecule.	2+1	

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Draw and label Endoplasmic reticulum, Golgi apparatus and Mitochondria. Also outline one role of each in a cell.	2+2 +2	OR	How does Karyokinesis occur in mitosis in a cell? Explain in detail. Also draw labelled diagram of the process.	4+2
Q.4	Describe the steps involved in aerobic respiration. Also write chemical equations.	06	OR	Explain development of female gametophyte plants. How double fertilization occurs in plants? Also draw life cycle of a plant.	3+1 +2
Q.5	How is the scientific method used to solve a biological problem? Explain all steps in correct sequence with brief explanation of each involved in scientific method.	06	OR	What are enzymes? Explain any five characteristics of enzymes.	1+5



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SECTION – A (Marks 12)

Time allowed: 20 Minutes

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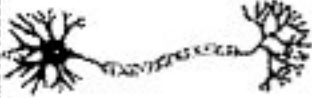
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Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Yellowing of leaves will occur in plants due to the deficiency of:	Magnesium	Calcium	Iron	Manganese	☐	☐	☐	☐
2. Strong wind increases transpiration rate mainly by:	Cooling leaves	Reducing light	Moving water vapours	Opening stomata	☐	☐	☐	☐
3. Tissue culture is an example of:	Sexual reproduction	Budding	Grafting	Cloning	☐	☐	☐	☐
4. In flowering plants, the zygote develops into:	Embryo	Ovule	Endosperm	Seed	☐	☐	☐	☐
5. Which cell is specialized for transmission of nerve impulses?	 Guard cell	 Neuron	 Root hair	 Amoeba	☐	☐	☐	☐
6. During a research project, scientists from different fields criticise each other's methods. This practice mainly helps to:	Delay results	Create conflict	Improve accuracy	Limit discussion	☐	☐	☐	☐
7. Development of dog breeds with different traits is an example of:	Natural selection	Genetic drift	Mutation	Selective breeding	☐	☐	☐	☐
8. If a DNA strand is having A-T-C-G-T nucleotide sequence, the mRNA formed from it will be:	U-A-G-C-A	T-A-G-C-A	T-U-G-C-U	T-U-C-G-U	☐	☐	☐	☐
9. If two different species are crossed artificially, they will produce:	Fertile offsprings	Sterile offsprings	Identical offsprings	Offspring similar to one parent	☐	☐	☐	☐
10. The longest phase of the cell cycle is usually:	M phase	G ₂ phase	Interphase	S-phase	☐	☐	☐	☐
11. The tissue which supports and binds other stomach tissues together is:	Muscular	Epithelial	Nervous	Connective	☐	☐	☐	☐
12. An organic molecule permanently attached to enzyme is a/an:	Activator	Prosthetic group	Coenzyme	Cofactor	☐	☐	☐	☐



BIOLOGY SSC-I

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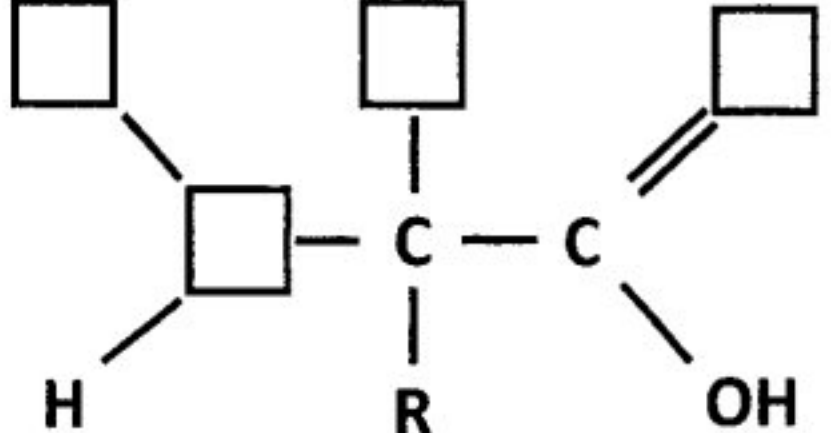
Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

Q.2 Attempt the following parts.

(10 x 3 =30)

(i)	Compare one structural advantage of epidermal cell in plants and red blood cell in animal that help each to perform its specific function.	1.5 +1.5	OR	State two differences between natural and artificial vegetative propagation in plants. And give one advantage of asexual reproduction.	2+1
(ii)	Differentiate between competitive and non-competitive inhibition.	03	OR	Why is meiosis important for producing variation in organisms? Explain briefly.	03
(iii)	Briefly describe Taxonomy, Immunology and Morphology with one example of each.	1x3	OR	Write any three differences between prophase of mitosis and prophase I of meiosis.	1x3
(iv)	Write about any three osmotic adjustments in xerophytes.	03	OR	State two dietary sources of lipids. Why are lipids important? Explain briefly.	2+1
(v)	Complete given structure of an amino acid with correct element / group. Also write name of functional groups found in an amino acid. 	2+1	OR	Identify the event of meiosis shown in figure and briefly explain it. 	03
(vi)	Briefly explain theory of natural selection by Darwin evolution.	03	OR	Briefly describe Computational Biology and Biochemistry give one example of each.	1.5 +1.5
(vii)	Differentiate between sucrose and maltose. (Any three differences)	1x3	OR	How is parthenogenesis different form of reproduction? Briefly explain with one example.	1+2
(viii)	Give at least two advantages and one disadvantage of Transpiration.	03	OR	How are mesophyll tissues important in a leaf due to their structure and function.	1.5 +1.5
(ix)	Name the groups included in domain Eukarya. Also write two features of this domain.	1+2	OR	Name any three tissues that are present in stomach. Give one function of each.	03
(x)	How vestigial organs and selective breeding provide support as evidences of evolution?	1.5 +1.5	OR	Identify the highest and lowest taxonomic rank used in modern classification. Also describe these briefly.	1.5 +1.5

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Explain light dependent reactions of photosynthesis. Also draw the diagram.	5+1	OR	Explain vegetative propagation in plants through Bulb, Rhizome and Tubers. Also draw diagrams.	2+2 +2
Q.4	How is the scientific method used to solve a biological problem? Explain all steps in correct sequence with brief explanation of each involved in scientific method.	06	OR	Describe the effect of temperature and pH on enzyme activity. (Explaining why enzymes work best under specific conditions)	3+3
Q.5	Draw and label nucleus. Explain the functions of different parts of nucleus.	2+4	OR	How does cytokinesis differ in plants and animal cell? Explain. Also draw the diagrams of both.	3+3